Xiang Zheng

Contact Information

Lau Ming Wai Academic Building, 8/F, City University of Hong Kong, Kowloon, Hong Kong SAR, China

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Webpage: https://x-zheng16.github.io
GitHub: https://github.com/x-zheng16
Google Scholar: 977 citations, h-index 11

Research Interests

Reinforcement Learning, Trustworthy AI, Agentic AI, Embodied AI, Trustworthy Recursive Self-Improvement, Agentic Red Teaming, AI for Science

Education

City University of Hong Kong **2020–2024**
Ph.D. in Computer Science, Hong Kong SAR, China
Supervisor: Prof. Cong Wang
Fields: Reinforcement Learning, Trustworthy AI
GPA: 4.00/4.00

Tsinghua University **2016–2019**
M.S. in Control Science and Engineering, Beijing, China
Supervisor: Prof. Tao Zhang
Fields: Reinforcement Learning, Intelligent Robotics
GPA: 3.55/4.00

Beihang University **2012–2016**
B.S. in Automation, B.S. in Mathematics (Dual Degree), Beijing, China
Selected into the Shen Yuan Honors College (top 0.5%, 1 in 200)
GPA: 3.74/4.00

Research & Work Experience

Hong Kong Institute of AI for Science (HKAI-Sci), City University of Hong Kong **2026–present**
Research Assistant Professor, Hong Kong SAR, China
Topic: Self-evolving science agents and trustworthy AI.
– Chief Scientist of SciencePal 2.0

City University of Hong Kong **2024.10–2026.03**
Postdoctoral Fellow, Hong Kong SAR, China
Supervisor: Prof. Cong Wang
Topic: Reinforcement learning for trustworthy and embodied AI.

Xi'an Jiaotong University **2022.09–2023.08**
Visiting Researcher, Xi'an, China
Supervisor: Prof. Chao Shen
Topic: Adversarial machine learning.

Xi'an Jiaotong University **2019.11–2020.02**

Visiting Researcher, Xi'an, China
Supervisor: Prof. Chao Shen and Xingjun Ma
Topic: Adversarial machine learning.

National Institute of Informatics **2018.02–2018.05**

Research Intern, Tokyo, Japan
Supervisor: Prof. Tetsunari Inamura
Topic: Intelligent robotics.

The University of New South Wales **2016.02–2016.06**

Research Intern, Sydney, Australia
Supervisor: Prof. Elias Aboutanios
Topic: Spacecraft de-orbiting control.

Publications

Conference (∗ for co-first author, † for corresponding author.)

C-1. Hanxun Huang, Yutao Wu, Qizhou Wang, Silvia Montaña-Niño, Yige Li, **Xiang Zheng**, Elif Buse Doyuran, Phoebe Matich, Xiao Liu, Xingjun Ma, Sarah Erfani, Christopher Leckie, "VEX-Bench: Benchmarking Verification Complexity of LLM-Generated Misinformation," in NeurIPS E&D Track, 2026.

C-2. Yutao Wu, Xiao Liu, Hanxun Huang, Yige Li, **Xiang Zheng**, Yifeng Gao, Cong Wang, Bo Li, Xingjun Ma, Yu-Gang Jiang, "Internal Safety Collapse in Frontier Large Language Models," in NeurIPS, 2026.

C-3. Zhouyuan Ma, Yutao Wu, Hanxun Huang, **Xiang Zheng**, Xiao Liu, Yixin Cao, Zuxuan Wu, Xingjun Ma, Yu-Gang Jiang, "HarmProfile: Characterizing Harmful Distributions in Frontier LLMs," in EMNLP Findings, 2026.

C-4. Xutao Mao, Liangjie Zhao, Tao Liu, **Xiang Zheng**†, Hongying Zan, Cong Wang†, "STARE: Step-wise Temporal Alignment and Red-teaming Engine for Multi-modal Toxicity Attack," in ICML, 2026.

C-5. **Xiang Zheng**, Yutao Wu, Hanxun Huang, Yige Li, Xingjun Ma†, Bo Li, Yu-Gang Jiang, Cong Wang†, "Just Ask: Curious Code Agents Reveal System Prompts in Frontier LLMs," in ICML, 2026.

C-6. Zilong Wang, **Xiang Zheng**†, Xiaosen Wang, Bo Wang, Xingjun Ma†, Yu-Gang Jiang, "GenBreak: Red Teaming Text-to-Image Generators Using Large Language Models," in CVPR, 2026.

C-7. Zonghuan Xu, Jiayu Li, Yunhan Zhao, **Xiang Zheng**†, Xingjun Ma†, Yu-Gang Jiang, "DropVLA: An Action-Level Backdoor Attack on Vision-Language-Action Models," in IROS, 2026.

C-8. Yunhan Zhao, **Xiang Zheng**†, Lin Luo, Yige Li, Xingjun Ma†, Yu-Gang Jiang, "BlueSuffix: Reinforced Blue Teaming for Vision-Language Models Against Jailbreak Attacks," in ICLR, 2025.

C-9. **Xiang Zheng**, Longxiang Wang, Yi Liu, Xingjun Ma, Chao Shen, Cong Wang†, "CALM: Curiosity-Driven Auditing for Large Language Models," in AAAI, 2025.

C-10. **Xiang Zheng**, Xingjun Ma, Chao Shen, Cong Wang†, "Constrained intrinsic motivation for reinforcement learning," in IJCAI, 2024.
Acceptance rate 791/5651 = 14.00%.

C-11. **Xiang Zheng**, Xingjun Ma, Shengjie Wang, Xinyu Wang, Chao Shen, Cong Wang†, "Toward evaluating robustness of reinforcement learning with adversarial policy," in DSN, 2024.
Acceptance rate 42/203 = 20.69%.

C-12. Shengjie Wang, Fengbo Lan, **Xiang Zheng**, Yuxue Cao, Oluwatosin Oseni, Haotian Xu, Tao Zhang, Yang Gao, "A policy optimization method towards optimal-time stability," in CoRL, 2023.

C-13. Shengjie Wang, **Xiang Zheng**, Yuxue Cao, Tao Zhang, "A multi-target trajectory planning of a 6-DoF free-floating space robot via reinforcement learning," in IROS, 2021.

C-14. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "An end-to-end trajectory planning strategy for free-floating space robots," in CCC, 2021.

C-15. Shihao Zhao, Xingjun Ma, **Xiang Zheng**, James Bailey, Jingjing Chen, Yu-Gang Jiang, "Clean-label backdoor attacks on video recognition models," in CVPR, 2020.
Acceptance rate 1470/6656 = 22.09%.

C-16. **Xiang Zheng**, Ziwei Wang, Tao Zhang, "Robust finite-time attitude tracking control for nonlinear quadrotor with uncertainties and delays," in ROBIO, 2017.

Journal

J-1. Yunhan Zhao, **Xiang Zheng**, Yige Li, Xingjun Ma†, "Defense-to-attack: Bypassing weak defenses enables stronger jailbreaks in Vision-Language Models," Pattern Recognition, 2026, art. no. 113805.
Impact Factor 7.6; JCR Rank Q1.

J-2. Yi Liu, **Xiang Zheng**, Chengjun Cai, Xingliang Yuan, Cong Wang†, "Adapting Large Language Models for Encrypted Traffic Analysis Services: An Efficient Realization with Mixture of LoRA Experts," IEEE Transactions on Services Computing, vol. 19, no. 2, pp. 906–919, 2026.
Impact Factor 5.5; JCR Rank Q1.

J-3. **Xiang Zheng**, Xingjun Ma, Wei-Bin Lee, Cong Wang†, "OpenRedRL: A Light-Weight Benchmark for Reinforcement Learning-Based Red Teaming," Frontiers of Computer Science, 2026.
Impact Factor 4.6; JCR Rank Q2.

J-4. Xingjun Ma, Yifeng Gao, Yixu Wang, . . . , **Xiang Zheng**, . . . , Yu-Gang Jiang (48 authors), "Safety at Scale: A Comprehensive Survey of Large Model and Agent Safety," Foundations and Trends in Privacy and Security, vol. 8, no. 3-4, pp. 254–469, 2026.

J-5. Yuxue Cao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Xinru Xie, Lei Liu, "Reinforcement learning with prior policy guidance for motion planning of dual-arm free-floating space robot," Aerospace Science and Technology (AST),

vol. 136, pp. 108098, 2023.
Impact Factor 5.6; JCR Rank Q1 (3/34).

J-6. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "A learning system for motion planning of free-float dual-arm space manipulator towards non-cooperative object," Aerospace Science and Technology (AST), vol. 131, pp. 107980, 2022.
Impact Factor 5.6; JCR Rank Q1 (3/34).

J-7. Shengjie Wang, Yuxue Cao, **Xiang Zheng**, Tao Zhang, "Collision-free trajectory planning for a 6-DoF free-floating space robot via hierarchical decoupling optimization," IEEE Robotics and Automation Letters (RA-L), vol. 7, no. 2, pp. 4953–4960, 2022.
Impact Factor 5.2; JCR Rank Q2 (10/30).

PREPRINTS

1. Xutao Mao, Liangjie Zhao, **Xiang Zheng**†, Cong Wang†, "Practice Makes Unsafe: Skill Misevolution in Self-Improving LLM Agents," arXiv preprint arXiv:2608.12851.

2. Xutao Mao, **Xiang Zheng**†, Cong Wang†, "Agent Hacks Agent: Autoresearch for Production-Agent Red-Teaming," arXiv preprint arXiv:2607.11698.

3. Xutao Mao, Liangjie Zhao, Leyao Wang, Rui Qian, Qiang Huang, Wentao Wang, Bo Han, **Xiang Zheng**†, Cong Wang†, "Agents Don't Just Agree, They Remember: Benchmarking Persistent Sycophancy in Stateful Personal Agents," arXiv preprint arXiv:2607.10526.

4. Yige Li, Jun Sun, Wei Zhao, Zhe Li, Yutao Wu, Hanxun Huang, **Xiang Zheng**, Xingjun Ma, "When Medical Safety Alignment Fails: A Benchmark for Evaluating LLMs on High-Risk Medical Queries," arXiv preprint arXiv:2606.28332.

5. Xiao Li*, **Xiang Zheng***, Yifeng Gao, ..., Xingjun Ma†, Yu-Gang Jiang† (38 authors), "Safety in Embodied AI: A Survey of Risks, Attacks, and Defenses," arXiv preprint arXiv:2605.02900.

6. Longxiang Wang, **Xiang Zheng**†, Xuhao Zhang, Yao Zhang, Ye Wu, Cong Wang†, "OptiLeak: Efficient Prompt Reconstruction via Reinforcement Learning in Multi-tenant LLM Services," arXiv preprint arXiv:2602.20595.

7. Zixing Chen, Yifeng Gao, Li Wang, Yunhan Zhao, Yi Liu, Jiayu Li, **Xiang Zheng**, Zuxuan Wu, Cong Wang, Xingjun Ma, Yu-Gang Jiang, "HazardArena: Evaluating Semantic Safety in Vision-Language-Action Models," arXiv preprint arXiv:2604.12447.

8. Zonghuan Xu, **Xiang Zheng**, Yutao Wu, Xingjun Ma, "When Direct Prediction Fails: Evidence from LLM-Based Misinformation Risk Evaluation," arXiv preprint arXiv:2604.06820.

9. Ming Wen, Yuxuan Liu, Kun Yang, Yunhao Feng, Zhuoer Xu, Yuhao Sun, Shiwen Cui, **Xiang Zheng**, Yi Liu, Xingjun Ma, Yu-Gang Jiang, "Constitutional On-Policy Safe Distillation," arXiv preprint arXiv:2606.03089.

10. Jiayu Li, Yunhan Zhao, **Xiang Zheng**, Zonghuan Xu, Yige Li, Xingjun

Ma, Yu-Gang Jiang, "AttackVLA: Benchmarking Adversarial and Backdoor Attacks on Vision-Language-Action Models," arXiv preprint arXiv:2511.12149.

11. Yutao Wu, Xiao Liu, Yinghui Li, Yifeng Gao, Yifan Ding, Jiale Ding, **Xiang Zheng**, Xingjun Ma, "ADMIT: Few-shot Knowledge Poisoning Attacks on RAG-based Fact Checking," arXiv preprint arXiv:2510.13842.

12. Jiale Ding, **Xiang Zheng**†, Yutao Wu, Cong Wang, Wei-Bin Lee, Ling Pan, Xingjun Ma†, Yu-Gang Jiang, "RedTopic: Toward Topic-Diverse Red Teaming of Large Language Models," arXiv preprint arXiv:2507.00026.

13. Yuxue Cao, Wenhao Zhao, Shengjie Wang, **Xiang Zheng**, Wenke Ma, Ziwei Wang, Tao Zhang, "A Learning-based Adaptive Compliance Method for Symmetric Bi-manual Manipulation," arXiv preprint arXiv:2303.15262.

Grants & Projects

Independent Grants as PI or Co-PI

Theme-based Research Scheme (TRS), Research Grants Council of Hong Kong **2027/28**

Co-PI; In preparation.

Co-written Grants

Foxconn-CityUHK Joint Research Centre **2027**

2027 Red Teaming Research Proposal (HKD 500K requested)
Proposal Co-writer; Submitted.

Theme-based Research Scheme (TRS), Research Grants Council of Hong Kong **2026/27**

SynapseCity: AI-Driven Urban Emergency Response and Rescue HKD 55.691M
Proposal Writer and Multi-Agent Prototype Contributor; Awarded.

Research Projects

Foxconn-CityUHK Joint Research Centre **2026**

Red Teaming Technology Research for Intelligent Agents HKD 500K
Participating Researcher; In progress.

Foxconn-CityUHK Joint Research Centre **2025**

Intrinsically Motivated RFT for Red Teaming Black-Box LLMs HKD 500K

CRF, No. 8730083 **2023–present**

Enabling Metadata-Private and Accountable Networks at Scale HKD 7.52M

NSFC, No. 9054034 **2022–present**

Towards Secure and Privacy-enhanced Machine Learning as a Service CNY 1M

RFS, No. 9062004 **2022–present**
Building Privacy-Assured and Scalable Encrypted Databases With Secure Enclave HKD 5.16M

CAST **2017–2019**
Learning-Based Control Technology CNY 1.8M

Invited Talks

Dalian, Medical Large Model Agent Forum, CHIP 2026 **2026.10.31**
SciencePal 2.0 Research Agent

Chengdu, Auto-Research Forum, CNCC 2026 **2026.10**
SciencePal 2.0

Online, Southern University of Science and Technology (Prof. Jianyu Niu's Group) **2026.09.03**
Trustworthy Recursive Self-Improvement

Hong Kong, Victoria Harbour Academic Salon **2026.08.27**
Towards Trustworthy Agents: From Red Teaming to Trustworthy Recursive Self-Improving

Dongguan, CityUHK Joint Summer School on AI for Science **2026**
Hands-on with Claude Code: Frontier Code Agents for Research and Browser/Computer Use

Beijing, Tencent Cloud AI Industry Applications Summit **2026**
SciencePal 2.0: Your Self-Evolving Science Agent Team

Hong Kong, HKAI-Sci, CityUHK **2026**
Build Your Own Self-Evolving Science Agent with SciencePal 2.0

Shenzhen, ByteDance **2026**
Agentic Red Teaming and Blue Teaming

Hong Kong, College of Computing at CityUHK (with Prof. Cong Wang) **2026**
Is OpenClaw All You Need?

Online, Lancaster University (Prof. Ziwei Wang's Group) **2026**
From Robot Learning to Self-Evolving OpenClaw: Building and Securing AI Agent Systems

Hong Kong, HKAI-Sci, CityUHK **2026**
Harness Engineering

Online, Institute of Trustworthy Embodied AI at Fudan (Prof. Xingjun Ma's Group) **2026**

Reinforcement Learning for Agent Safety Evaluation

Xi'an, CCF YOCSEF **2025**

Towards Robust Embodied AI: Skill Discovery and Red Teaming

Shenzhen, Southern University of Science and Technology (Prof. Jianyu Niu's Group) **2025**

Reinforcement Learning-Based Adversarial Safety Evaluation and Defense Enhancement for Large Language Models

Xi'an, CCF YOCSEF **2025**

Reinforcement Fine-Tuning for Large Model Red-/Blue-Teaming

Xi'an, School of Computer Science at Northwestern Polytechnical University (Prof. Helei Cui's Group) **2024**

Curiosity-Driven Auditing for LLMs

Xi'an, School of Computer Science at Northwestern Polytechnical University (Prof. Helei Cui's Group) **2024**

Efficient Intrinsically Motivated Adversarial Policy Learning

Online, Griffith University (Prof. Leo Yu Zhang's Group) **2024**

Intrinsically Motivated Adversarial Policy

Melbourne, School of Computing and Information Systems at the University of Melbourne (Prof. Xingliang Yuan's and Prof. Feng Liu's Groups) **2024**

Towards Efficient Evasion Attacks Against RL

Awards & Honors

ICML Gold Reviewer Award **2026**

Top 25%

IJCAI Travel Grant **2024**

IJCAI Organization and AIJ Division
USD 500

International Conference Grant **2024**

City University of Hong Kong
HKD 10,000

DSN Student Travel Grant **2024**

DSN Student Travel Awards Committee
USD 1,500

Research Activities Fund **2022–2023**

City University of Hong Kong
HKD 96,000

Institutional Research Tuition Grant **2020–2024**
City University of Hong Kong
HKD 168,384

CityUHK Presidential PhD Scholarship **2020–2024**
2/73
HKD 1,353,120

NII MOU Research Activities Grant **2018**
1/106
National Institute of Informatics, Japan
JPY 68,400

CSC Scholarship **2016**
China Scholarship Council
AUD 6,400

Stars of Advanced Engineering **2015**
Shen Yuan Honors College, Beihang University
Top-2 of 50 honors students
Highest honor at the Honors College

Teaching

CS5293, Topics on Information Security
CS4394, Information Security and Management
CS4293, Topics in Cybersecurity
CS5294, Information Security Technology Management
CS6290, Privacy-enhancing Technologies
CS2310/CS2311, Computer Programming

Mentored Students

City University of Hong Kong
Longxiang Wang (PhD), RL, AI for Privacy, Agent Safety
Pinsheng Xu (PhD), Multi-Agent System Safety
Xutao Mao (PhD), RL, VLM Safety, Agent Safety [ICML'26]
Chenghao Yao (MS), RL, Unlearning
Shuxuan Lye (MS), GUI Agent Safety
Xiao Hu (MS), RL, Agent-as-Judge
Cheng Gui (RA), Recursive Self-Improvement Safety

Fudan University
Ruofan Wang (PhD), RL, Red Teaming for VLM
Zilong Wang (PhD), RL, Red Teaming for Text-to-Image Model [CVPR'26]
Xiao Li (MS), Multi-Agent RL Safety
Yunhan Zhao (MS), RL, Blue Teaming for VLM [ICLR'25; PR'26; UIUC PhD '26]
Zixing Chen (MS), RL, VLA Safety
Jiayu Li (MS), VLA Safety
Jiale Ding (UG), RL, LLM Safety [HKUST PhD '26 (HK PhD Fellowship)]
Zonghuan Xu (UG), VLA Safety, Misinformation [IROS'26]

Tsinghua University

Shengjie Wang (MS), RL, Safe RL, Robot Learning, Space Robot Control [IROS'21; RA-L'22; AST'22/'23; CoRL'23; DSN'24; Tsinghua PhD '22]

Institute of Automation
Yuxue Cao (MS), RL, Robot Learning, Space Robot Control [IROS'21; RA-L'22; AST'22/'23; CoRL'23; Siemens]

Northwestern Polytechnical University
Siyuan Zhang (MS), Agent for Crowdsensing

Professional Activity

Program Committee Member
NeurIPS 2026
ICML 2026
ICLR 2025, 2026, 2027
CVPR 2026
ECCV 2026
EMNLP 2026
AAAI 2025, 2026, 2027
MM 2025, 2026
ICRA 2026

Organization
Special Session Organizer: PRCV 2026

Journal Reviewer
CSUR, TPAMI, TDSC, TSC, TC, TMLR, IoT-J

External Conference Reviewer
ACL 2026
NeurIPS 2025
ICNP 2025
ESORICS 2022
AsiaCCS 2022
RAID 2021